

# **Training Method Efficiency & Variability Analysis**

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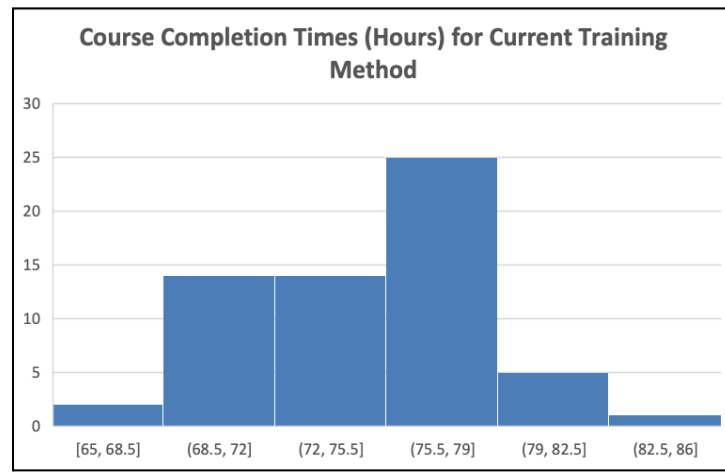
Data Analysis Project

*Microsoft Excel | Statistical Analysis*

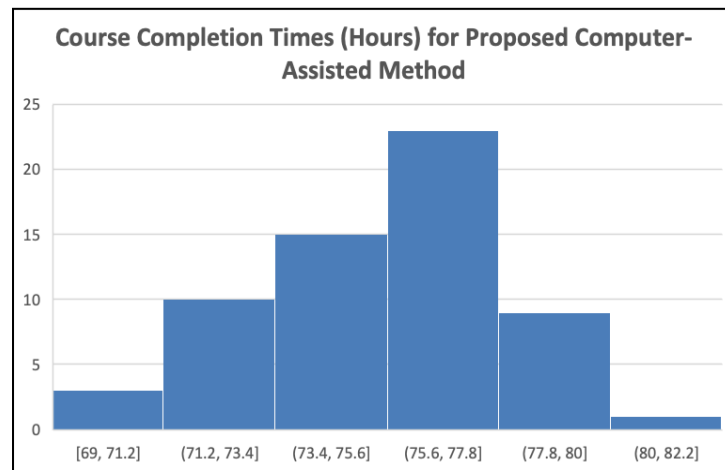
## Summary Statistics

The mean course completion times for the Current Training Method are 75.0656 hours, while for the New Proposed Method are 75.4262 hours. While the mean times are nearly equal, slightly more students take longer using the new proposed method. Secondly, the median for both methods is 76 hours. Thirdly, the mode for both methods is 76 hours, meaning that the most hours that students took the most time on were 76 hours.

In addition, we can visualize the data by creating histogram charts for each method.



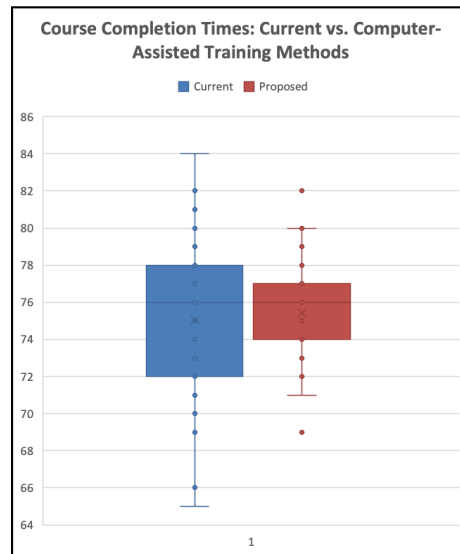
*Figure 1. Course Completion Times for Current Training Method Histogram*



*Figure 2. Course Completion Times for Proposed Computer-Assisted Method Histogram*

By analyzing the histograms in Figures 1 and 2, we can see that both methods are slightly left-skewed. Most students' completion times were in the range of (75.5,79] for both methods, while the least students' times were in the upper and lower ranges, most likely illustrating outliers.

We can also compare both methods by creating box and whisker plots of each method and placing them side-by-side.



*Figure 3. Course Completion Times for Current Training Method Box-and-Whisker Plot*

Figure 3 illustrates that we can see that the new proposed method is less spread out than the current method. This exemplifies that while both methods have similar means and modes, the new proposed method allows students to have more similar completion times using the computerized method.

Overall, both methods have similar means and modes, but differ in their variability. The Current Training Method shows a wider range of times, meaning that students' completion rates vary more. On the other hand, the Proposed Computer-Assisted Method has a smaller spread, showing that students tend to finish the course within a more similar time frame. This suggests that while the average completion time does not change much, the new computerized method may help create a more consistent and efficient learning experience for students.

## Hypothesis Testing

Using the data, we can conduct a hypothesis test on the difference between the population means for the two methods.

Let

- $\mu_1$  = mean completion time (current programmed-text method)
- $\mu_2$  = mean completion time (computer-assisted method)

We test whether the new method reduces completion time:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

Significance level:  $\alpha = 0.05$

Standard deviation:

$$s_1 = 3.945$$

$$s_2 = 2.506$$

Standard error of difference:

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{(3.945)^2}{61} + \frac{(2.506)^2}{61}} = 0.598$$

Two-Sample T-Test:

$$t = \frac{(75.0656 - 75.4262) - 0}{\sqrt{\frac{(3.945)^2}{61} + \frac{(2.506)^2}{61}}} = -0.603$$

Degrees of freedom:

$$df = \frac{\left(\frac{(3.945)^2}{61} + \frac{(2.506)^2}{61}\right)^2}{\frac{1}{61-1}\left(\frac{(3.945)^2}{61}\right)^2 + \frac{1}{61-1}\left(\frac{(2.506)^2}{61}\right)^2} = 101.642$$

P-value:

$$p \approx 0.55$$

$$p > \alpha$$

**Fail to reject  $H_0$**

We fail to reject the null hypothesis because there is no statistically significant difference between the mean training times for the current and proposed methods. The small observed difference (about 0.36 hours) could easily have occurred due to random sampling variation.

## Test for Equality of Variances

Standard deviation:

$$s_1 = \sqrt{\frac{\sum (x_i - 75.0656)^2}{61-1}} = 3.945$$

$$s_2 = \sqrt{\frac{\sum (x_i - 75.4262)^2}{61-1}} = 2.506$$

Variance:

$$s_1^2 = 15.563$$

$$s_2^2 = 6.280$$

Hypothesis:

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Significance level:  $\alpha = 0.05$

Test statistic:

$$F = \frac{15.563}{6.280} = 2.478$$

Degrees of freedom:

$$df_1 = 61 - 1 = 60$$

$$df_2 = 61 - 1 = 60$$

$$F_{0.975,60,60} = 1.667$$

$$F_{0.025,60,60} = 0.5999$$

$$F > 1.667$$

Reject  $H_0$

There is sufficient evidence at the 5% significance level to conclude that the population variances for the two training methods are not equal. The current method shows significantly more variation in completion times.

## Hypothesis Tests Conclusion

The hypothesis tests show that there is no statistically significant difference between the mean completion times for the two methods. The P-value of 0.55 is greater than the 5% significance level, so we fail to reject the null hypothesis. This means that, on average, students complete the course in roughly the same amount of time under both methods. On the other hand, the F-test on variances found a significant difference, which indicates that the new proposed method has less variability in completion times. Students working through the computer-based system tend to finish within a more consistent time frame, while the current text-based method shows wider variation.

In conclusion, although the average completion times are similar, the new computer-assisted method provides more uniform results. This consistency may make scheduling and pacing easier

for instructors and could improve training efficiency. Therefore, the computer-assisted method is the better approach.

### **Recommendations and Future Data Collection**

Before making a final decision on choosing the training method to be used in the future, it would be useful to collect additional data. For example, the Air Force could measure not only completion time but also how well students retain the material and perform on later assessments. Gathering feedback from instructors and students about usability and engagement could also provide valuable insight. Additionally, they could repeat the study with a larger sample group to confirm that the results are consistent. Finally, tracking long-term outcomes, such as performance in subsequent courses or field tasks, would illustrate whether one method leads to better overall learning. Collecting this extra information would support a more confident and evidence-based decision about which training method is most effective.